SAECOIN

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Name: SAECOIN
Ticker Symbol: SAC (Provisional)
Document: Product Requirements Document
Version: 1.0
Project Type: Independent Layer-1 Blockchain
Consensus Mechanism: Proof of Work (PoW)
Development Status: Planning
Target Platform: Windows, Linux, Android
License: Open Source (Final License To Be Determined)

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1. PROJECT OVERVIEW

SAECOIN is an independent Layer-1 cryptocurrency project designed to operate on its own decentralized blockchain network.

SAECOIN will have its own:

- Blockchain
- Native Cryptocurrency
- Mining System
- Full Node Software
- Peer-to-Peer Network
- Transaction System
- Wallet Applications
- Block Explorer
- Mining Pool Infrastructure

SAECOIN will not depend on Ethereum, BNB Chain, Solana, or another existing blockchain for its native currency.

The primary objective is to create a decentralized digital currency that allows users worldwide to send, receive, hold, and mine SAECOIN.

The network will use Proof of Work consensus.

CPU and GPU mining support is a core project requirement.

The protocol must be independently verifiable and open source.

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2. PROJECT VISION

The vision of SAECOIN is to create an accessible, secure, transparent, and decentralized cryptocurrency network.

The project aims to:

1. Allow ordinary users to participate in mining.
2. Support CPU and GPU mining.
3. Provide independent blockchain infrastructure.
4. Enable direct peer-to-peer transactions.
5. Maintain a predictable monetary supply.
6. Provide secure non-custodial wallets.
7. Allow anyone to operate a full node.
8. Maintain transparent public transactions.
9. Support future ecosystem expansion.
10. Minimize dependence on centralized infrastructure.

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3. CORE PROJECT IDENTITY

Property| Specification
Project Name| SAECOIN
Coin Name| SAECOIN
Ticker| SAC (Provisional)
Blockchain Type| Independent Layer-1
Consensus| Proof of Work
Mining| CPU + GPU
Maximum Supply| 21,000,000 SAC
Initial Block Reward| 50 SAC
Target Block Time| 10 Minutes
Halving Interval| 210,000 Blocks
Premine| 0
Decimal Precision| 8
Transaction Visibility| Public
Transaction Model| UTXO
Open Source| Yes

All monetary values must use integer arithmetic in the smallest denomination.

The maximum supply must never be exceeded.

The ticker symbol must be checked for conflicts before public exchange integration.

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4. PROJECT OBJECTIVES

4.1 Primary Objectives

The system must:

- Operate as an independent blockchain.
- Validate transactions without a central authority.
- Validate blocks using Proof of Work.
- Support CPU and GPU mining.
- Maintain decentralized peer-to-peer communication.
- Provide secure cryptocurrency wallets.
- Allow independent node operation.
- Maintain a public transaction ledger.
- Support blockchain synchronization.
- Prevent unauthorized coin creation.

4.2 Secondary Objectives

Future development may include:

- Mining Pool Software.
- Advanced Desktop Node Application.
- Android Wallet Improvements.
- Windows Wallet Improvements.
- Blockchain Analytics.
- Third-party Token Creation.
- Developer Infrastructure.
- Additional ecosystem services.

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5. TARGET USERS

SAECOIN is designed for the following user groups.

5.1 Regular Cryptocurrency Users

Users who want to:

- Create wallets.
- Receive SAC.
- Send SAC.
- Check balances.
- View transactions.

5.2 Miners

Users who want to:

- Mine SAC using CPUs.
- Mine SAC using GPUs.
- Participate in solo mining.
- Join mining pools.

5.3 Node Operators

Users who want to:

- Run full nodes.
- Verify transactions.
- Synchronize blockchain data.
- Support network decentralization.

5.4 Developers

Developers who want to:

- Inspect the source code.
- Build node software.
- Integrate wallet functionality.
- Develop future blockchain ecosystem components.

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6. BLOCKCHAIN REQUIREMENTS

6.1 Independent Blockchain

SAECOIN must have its own blockchain.

Required components:

- Genesis Block.
- Block Header.
- Block Hash.
- Previous Block Hash.
- Merkle Root.
- Timestamp.
- Nonce.
- Difficulty Target.
- Transaction Data.
- Block Validation.

Every full node must independently validate blocks according to the same consensus rules.

6.2 Genesis Block

The blockchain must begin with a predefined Genesis Block.

Genesis Block requirements:

- Unique Genesis Block Hash.
- Unique Network Identifier.
- Initial Difficulty.
- Initial Timestamp.
- Protocol Version.
- Initial Consensus Parameters.

The Genesis Block must be identical across all official network nodes.

The Genesis Block must not create an undisclosed premine.

6.3 Block Time

Target block interval:

10 Minutes.

The protocol must include automatic difficulty adjustment.

Difficulty adjustment parameters must be finalized in the Consensus Specification.

6.4 Blockchain Validation

Every full node must verify:

- Block structure.
- Proof of Work.
- Block hash.
- Difficulty target.
- Transaction signatures.
- Transaction inputs.
- Transaction outputs.
- Transaction fees.
- Block reward.
- Total monetary supply.
- Consensus rules.

Invalid blocks must be rejected.

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7. CONSENSUS REQUIREMENTS

SAECOIN will use Proof of Work.

The consensus system must:

1. Validate computational Proof of Work.
2. Enforce block reward rules.
3. Enforce monetary supply limits.
4. Prevent invalid transaction inclusion.
5. Validate block timestamps.
6. Support chain reorganization.
7. Resolve competing valid chains using a deterministic cumulative-work rule.
8. Protect against invalid block submissions.

The final mining algorithm will be selected after technical benchmarking.

CPU and GPU mining support must be evaluated together.

ASIC resistance is a design objective, not a guaranteed property.

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8. MINING SYSTEM REQUIREMENTS

The mining system must support:

8.1 CPU Mining

- Windows CPU Miner.
- Linux CPU Miner.
- Configurable CPU Threads.
- Mining Statistics.
- Hashrate Monitoring.
- Mining Reward Tracking.

8.2 GPU Mining

- Compatible GPU Mining Software.
- GPU Device Detection.
- GPU Hashrate Monitoring.
- Mining Performance Statistics.
- Temperature Monitoring where supported.

8.3 Mining Modes

- Solo Mining.
- Pool Mining.
- Local Test Mining.

8.4 Mining Reward

Initial Block Reward:

50 SAC.

Halving Interval:

210,000 Blocks.

Block rewards must follow deterministic consensus rules.

The reward schedule must use integer smallest-unit calculations to guarantee that the maximum supply is never exceeded.

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9. TRANSACTION SYSTEM

SAECOIN will use a UTXO-based transaction model.

The transaction system must support:

- Wallet-to-Wallet Transfers.
- Transaction Signatures.
- Transaction Verification.
- Input Validation.
- Output Validation.
- Transaction Fees.
- Mempool.
- Transaction Identifiers.
- Double-Spending Prevention.

All transactions must be publicly verifiable.

Private keys must never be transmitted as part of a transaction.

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10. TRANSACTION CONFIRMATION

The network will use six confirmations as the recommended high-confidence payment policy.

The wallet must display:

- Unconfirmed.
- 1 Confirmation.
- 2 Confirmations.
- 3 Confirmations.

- 4 Confirmations.
- 5 Confirmations.
- 6+ Confirmations.

The application must not describe six confirmations as an absolute guarantee against reorganization or attack.

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11. WALLET REQUIREMENTS

SAECOIN will provide:

1. Android Wallet.
2. Windows Wallet.

Both wallets must support:

- Create Wallet.
- Import Wallet.
- Wallet Backup.
- Wallet Restore.
- Receive SAC.
- Send SAC.
- QR Code.
- Transaction History.
- Balance Display.
- Address Management.
- Private Key Protection.

Wallets must be non-custodial.

Users must retain control of their private keys.

Private keys must be encrypted at rest.

Seed phrase generation and recovery must follow established cryptographic standards.

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12. NODE SOFTWARE REQUIREMENTS

SAECOIN must provide two node operation methods.

12.1 Desktop Node Application

Features:

- Install Node.
- Start Node.
- Stop Node.
- Blockchain Synchronization.
- Network Connection Status.
- Block Height.
- Peer Count.
- Node Statistics.
- Mining Status.

12.2 Command Line Interface

The CLI must support commands for:

- Starting a Node.
- Stopping a Node.
- Checking Node Status.
- Checking Blockchain Height.
- Listing Connected Peers.
- Creating Wallets.
- Checking Wallet Balances.
- Sending Transactions.
- Starting Mining.
- Checking Mining Statistics.

Exact commands will be defined in the CLI Specification.

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13. PEER-TO-PEER NETWORK

SAECOIN must operate through a decentralized P2P network.

Requirements:

- Node Discovery.
- Peer Connection.
- Block Propagation.
- Transaction Propagation.
- Blockchain Synchronization.
- Peer Validation.
- Network Reconnection.
- Protection Against Malicious Peers.
- Network Version Compatibility.

No single central server should be required for normal transaction validation or consensus.

Bootstrap nodes may be provided for initial peer discovery.

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14. BLOCK EXPLORER

A public Block Explorer must be developed.

Required features:

- Network Overview.
- Latest Blocks.
- Block Details.
- Transaction Details.
- Wallet Address Search.
- Transaction Search.
- Block Height Search.
- Mining Difficulty.
- Network Hashrate.
- Circulating Supply.
- Transaction Statistics.

The Explorer must obtain blockchain information from independently operating nodes or trusted indexing infrastructure.

The Explorer must not become a central authority for blockchain validation.

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15. MINING POOL

A mining pool system is planned.

Requirements:

- Miner Registration.
- Worker Management.
- Hashrate Statistics.
- Share Verification.
- Reward Calculation.
- Pool Statistics.
- Payout Management.

Pool accounting must be independently verifiable.

The mining pool must not be required for solo mining.

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16. FUTURE TOKEN ECOSYSTEM

Future SAECOIN development may support third-party token creation.

This feature is not required for the initial mainnet launch.

The token protocol must be designed separately from the native SAC currency.

Future requirements:

- Token Creation.
- Token Identification.
- Token Transfer.
- Token Supply Rules.
- Token Ownership.
- Token Transaction Verification.
- Explorer Integration.

The token system must not allow unauthorized modification of native SAC monetary rules.

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17. OPEN SOURCE REQUIREMENTS

SAECOIN will be an open-source project.

The source code should include:

- Blockchain Core.
- Node Software.
- Mining Software.
- Wallet Software.
- Explorer Software.
- Protocol Documentation.
- Build Instructions.
- Testing Instructions.

Source code must be publicly accessible.

A final open-source license must be selected before the public repository release.

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18. SECURITY REQUIREMENTS

Security is a critical requirement.

The system must include:

- Cryptographically Secure Private Keys.
- Digital Signature Verification.
- Double-Spending Protection.
- Consensus Validation.
- Block Validation.
- Transaction Validation.
- Replay Protection.
- Peer-to-Peer Network Protection.
- Secure Wallet Backup.
- Encrypted Key Storage.
- Denial-of-Service Mitigation.
- Safe Blockchain Reorganization Handling.

All critical cryptographic operations must use reviewed cryptographic libraries.

Custom cryptographic algorithms must not be invented without extensive expert review.

An independent security audit is required before mainnet launch.

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19. PERFORMANCE REQUIREMENTS

The system should aim for:

- Reliable blockchain synchronization.
- Efficient block validation.
- Efficient transaction propagation.
- Stable P2P connections.
- Low resource usage for ordinary nodes.
- CPU mining compatibility.
- GPU mining compatibility.
- Efficient blockchain storage.

The initial development environment must support local testing on Windows 11 with approximately 8 GB RAM.

Large-scale production requirements will be evaluated separately.

No unverified network throughput or transaction-per-second claims should be published.

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20. GOVERNANCE AND DECENTRALIZATION

The initial development team will coordinate development and testing.

However, the long-term objective is a decentralized network.

The protocol must not include an undisclosed centralized mechanism that allows the development team to arbitrarily:

- Create additional SAC.
- Modify user balances.
- Reverse valid transactions.
- Seize wallet funds.
- Change consensus rules unilaterally.

Protocol upgrades must use transparent versioning and publicly documented activation rules.

Network participants must independently verify consensus rules.

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21. DEVELOPMENT PHASES

Phase 1: Documentation

- PRD.
- App Flow.
- Technical Requirements.
- Blockchain Architecture.
- Consensus Specification.
- Mining Specification.
- Security Specification.

Phase 2: Blockchain Core

- Genesis Block.
- Block Structure.
- Hashing.
- Proof of Work.
- Block Validation.
- Difficulty Adjustment.

Phase 3: Transaction System

- UTXO.
- Transaction Creation.
- Transaction Signing.
- Transaction Validation.
- Mempool.

Phase 4: Node Network

- P2P Networking.
- Peer Discovery.
- Block Propagation.
- Synchronization.

Phase 5: Wallets

- Android Wallet.
- Windows Wallet.
- Wallet Backup.
- Send/Receive.

Phase 6: Mining

- CPU Miner.
- GPU Miner.
- Solo Mining.
- Mining Pool.

Phase 7: Explorer

- Block Explorer.
- Address Search.
- Transaction Search.
- Network Statistics.

Phase 8: Public Testnet

- Public Node Testing.
- Mining Testing.
- Transaction Testing.
- Security Testing.

Phase 9: Mainnet Preparation

- Independent Security Audit.
- Consensus Review.
- Supply Verification.
- Genesis Configuration.
- Public Release.

Phase 10: Mainnet Launch

- Public Network Activation.
- Independent Node Participation.
- Mining Activation.
- Wallet Release.

- Explorer Release.

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22. ACCEPTANCE CRITERIA

The initial SAECOIN blockchain implementation will be considered functionally complete only when:

1. Genesis Block is reproducible.
2. Multiple independent nodes can communicate.
3. Nodes independently validate blocks.
4. Proof of Work is verified.
5. Transactions can be created and signed.
6. Transactions can be broadcast.
7. Valid transactions can be included in blocks.
8. Invalid transactions are rejected.
9. Double spending is rejected.
10. Mining rewards follow the specified schedule.
11. Maximum supply cannot be exceeded.
12. Blockchain synchronization works.
13. Wallets can send and receive SAC.
14. CPU mining works.
15. GPU mining works on supported hardware.
16. Blockchain data can be independently verified.

Mainnet release additionally requires successful security review and public testnet evaluation.

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23. IMPORTANT DEVELOPMENT RULES

All developers and AI coding agents must follow these rules:

1. PRD is the primary product requirements document.
2. All consensus rules must be documented.
3. Never implement undocumented monetary changes.
4. Never introduce a premine.
5. Never modify maximum supply without an explicitly approved protocol revision.
6. Never store plaintext private keys.
7. Never use fake blockchain data in production.
8. Never claim successful mining without verified Proof of Work.
9. Never claim decentralization when the network depends on one central authority.
10. All critical code must have automated tests.
11. All protocol changes must be version-controlled.
12. Security-critical changes require additional review.

AI coding agents must inspect the relevant documentation before implementing any module.

If a requirement is ambiguous, the agent must document the ambiguity instead of inventing a consensus rule.

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24. FINAL PROJECT GOAL

The final objective is to launch SAECOIN as an independent, open-source, Proof of Work cryptocurrency network.

The network will provide:

- Independent Blockchain.
- Native SAC Currency.
- CPU and GPU Mining.
- Decentralized Full Nodes.
- Android Wallet.
- Windows Wallet.
- Desktop Node Software.
- Command Line Tools.
- Public Block Explorer.
- Mining Pool Infrastructure.
- Future Third-Party Token Support.

SAECOIN must prioritize security, transparency, decentralization, and long-term sustainability.

END OF DOCUMENT